Name	Zhoubin Li
Student ID	34906320
Studio Session	S15

Part A: URL of the Visualisation

Paste your visualisation URL in the empty space below.
Tip: Test your URL in incognito mode to ensure it is a working URL
https://zhoubin999.github.io/FIT3179_DV2_AUS_Radio/

Part B: Domain Description

Describe your domain chosen for DV II
My specific topic is how Australian radio is changing as streaming audio, podcasts and digital broadcasting become more widespread. The visualisation examines this transition through four connected dimensions: changes in audio consumption, differences across age groups and regions, the distribution of radio infrastructure, and station preferences in five metropolitan markets. Each dataset retains the population and time period defined by its source rather than treating all survey percentages as directly comparable.

Describe the Why and Who aspects of your DV II
Why:
To reveal patterns that national listening figures can hide—how streaming and radio coexist, which age groups and regions differ in their listening habits, where digital broadcasting infrastructure is concentrated, and how station preferences and rankings vary between cities. The goal is to communicate that radio's transformation is uneven rather than a simple replacement by streaming.

Who:
Australian listeners, the public and community or media organisations interested in changing audio habits. Broadcasting terminology and survey differences are explained in plain language.

Describe the "What" aspect of your DV II

The visualisation combines survey and radio-ratings tables from Edison Research (Infinite Dial Australia, 2017–2026), ACMA listening surveys, ACMA transmitter records, ABS population statistics, and GfK metropolitan radio ratings (Surveys 3–6, 2026).

Quantitative attributes include weekly reach, listening and ownership percentages, station listening shares, transmitter power and site counts. Categorical, temporal and spatial attributes include age group, region, listening service, station, program type, city, state, year, survey and geographic location. Derived fields include percentage-point differences, site density per 100,000 people, combined program-type shares, unrated listening shares, unweighted five-city averages and station rankings.

Describe the "How" aspect of your DV II

A findings-led narrative presents 13 charts across four analytical chapters and a concluding section. Line and bump charts reveal temporal trends and changes in station rankings, while dumbbell and diverging bar charts support comparisons between groups. Heatmaps highlight patterns through colour luminance, and mosaic charts, waffle charts, treemaps and donut glyphs communicate distributions and proportions. Radar charts compare listening contexts, while an alluvial diagram illustrates changes in average program-type shares across surveys.

Three distinct map idioms support different geographical tasks: proportional symbols show transmitter locations and magnitudes, choropleth shading reveals state-level infrastructure differences, and donut glyphs compare listening preferences across cities.

Visual marks include points, lines and areas, with position, length, area, angle, colour hue and luminance encoding quantitative and categorical attributes. Aggregation, filtering and juxtaposed views support comparisons at different levels of detail. Linked city selections, survey controls and map drill-downs enable interactive exploration, while findings-led headings, annotations and consistent typography guide readers through the story.

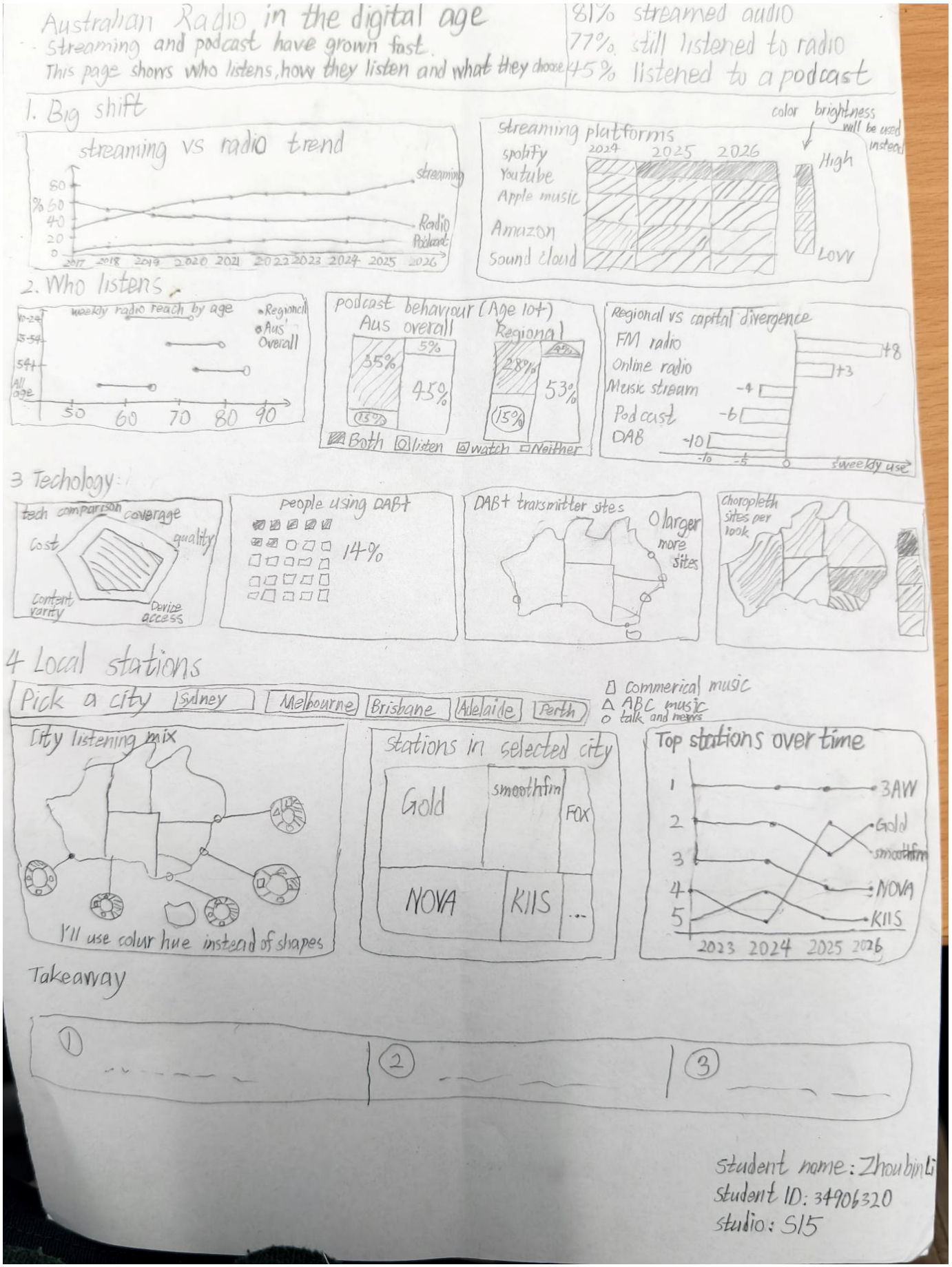
Part C: Sketch

A URL to a PDF document with an image of your sketch on your GitHub account.
Tip: Test your URL in incognito mode to ensure it is a working URL

https://github.com/Zhoubin999/FIT3179_DV2_AUS_Radio/blob/main/sketch.pdf

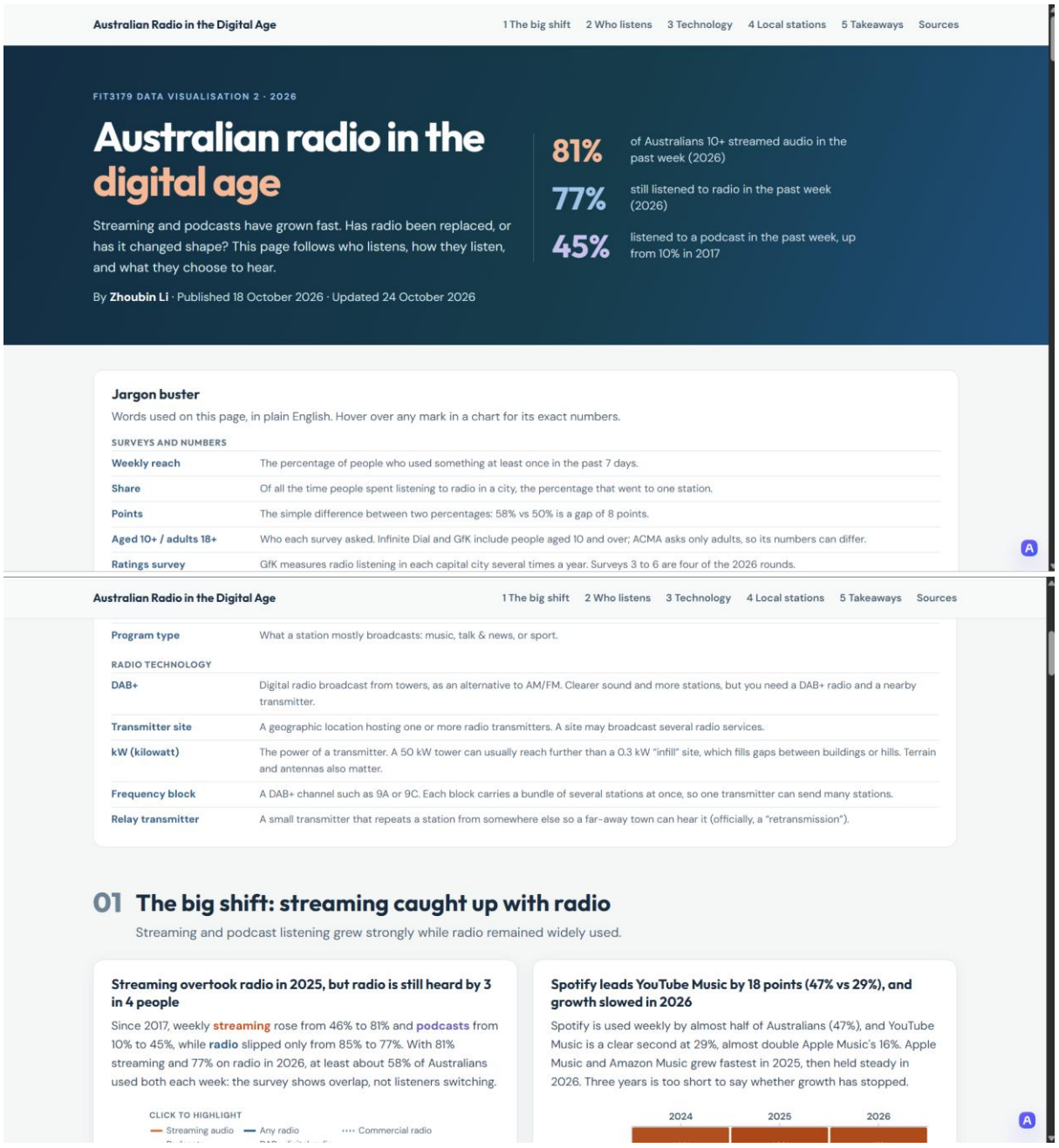
Paste a screenshot of your sketch for your DV II.
Note: Use the entire page here for the screenshot and ensure that it is clearly legible and visible

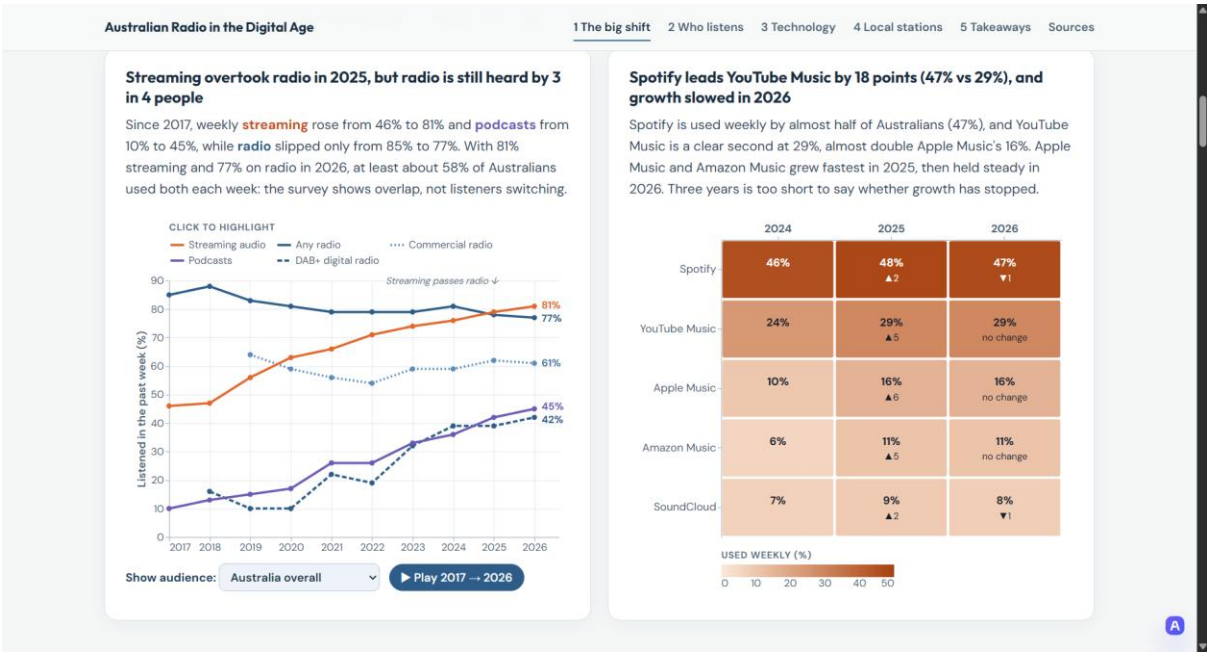
---

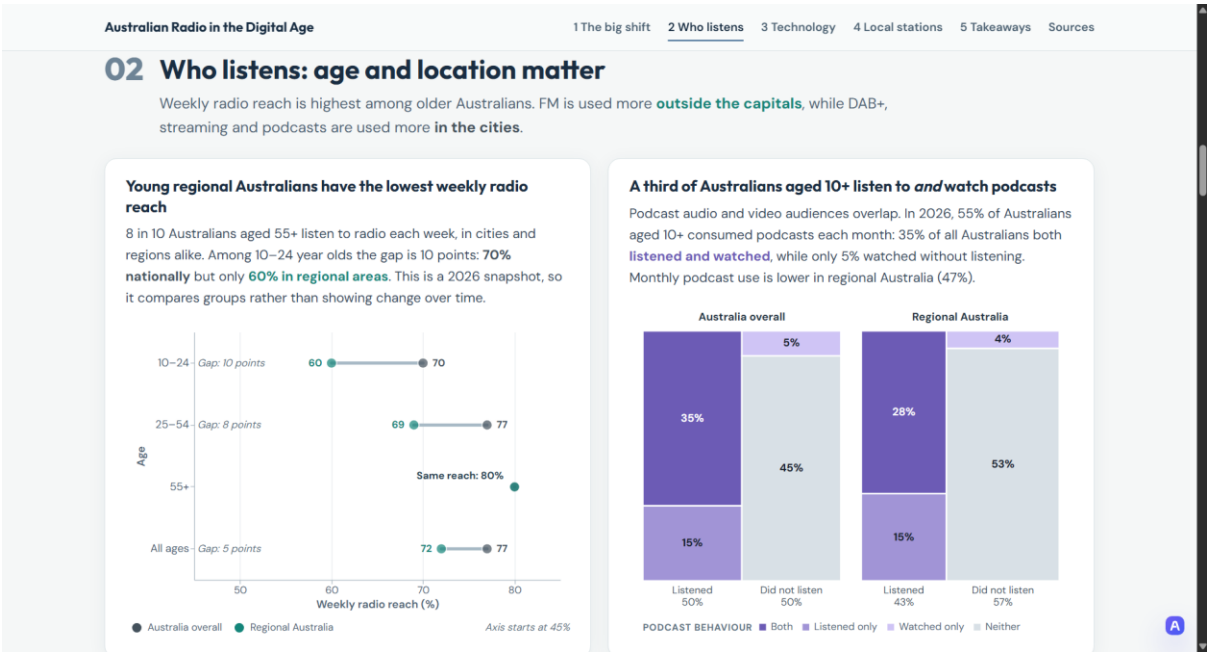

Australian Radio in the digital age
streaming and podcast have grown fast.
This page shows who listens, how they listen and what they choose
81% streamed audio
77% still listened to radio
45% listened to a podcast
1. Big shift
streaming vs radio trend
streaming
Radio
Podcast
streaming platforms
spotify
Youtube
Apple music
Amazon
Sound cloud
2024
2025
2026
color brightness will be used instead
High
Low
2. Who listens
weekly radio reach by age
Regional
Aus Overall
podcast behaviour (Age 10+)
Aus overall
Regional
35%
5%
45%
28%
53%
15%
Both
listen
watch
Neither
Regional vs capital divergence
FM radio
Online radio
Music stream
Podcast
DAB
+8
+3
-4
-6
-10
weekly use
3 Techology
tech comparison
coverage
quality
Device access
content variety
cost
People using DAB+
14%
DAB+ transmitter sites
larger more sites
choropleth sites per book
4 Local stations
Pick a city
Sydney
Melbourne
Brisbane
Adelaide
Perth
commerical music
ABC music
talk and news
City listening mix
I'll use colur hue instead of shapes
Stations in selected city
Gold
smoothfm
FOX
NOVA
KIIS
...
Top stations over time
3AW
Gold
smoothfm
NOVA
KIIS
2023
2024
2025
2026
Takeaway
1
2
3
Student name: Zhoubin Li
Student ID: 34906320
Studio: S15

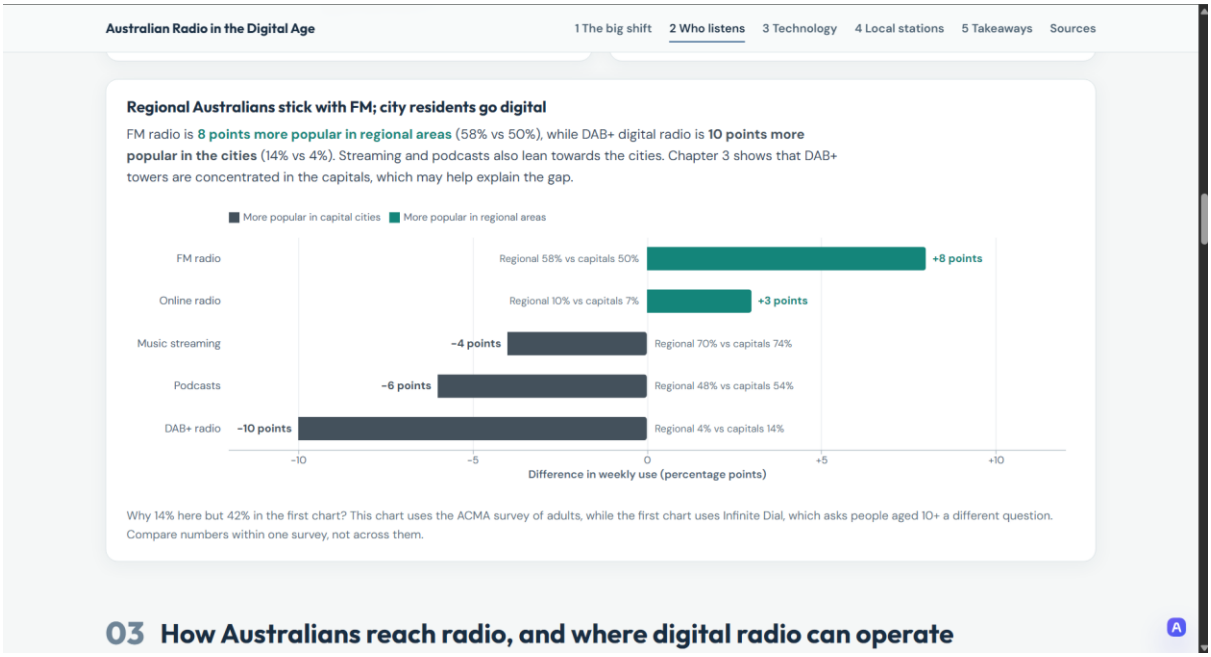
Part D: Full-size screenshot of your DV II

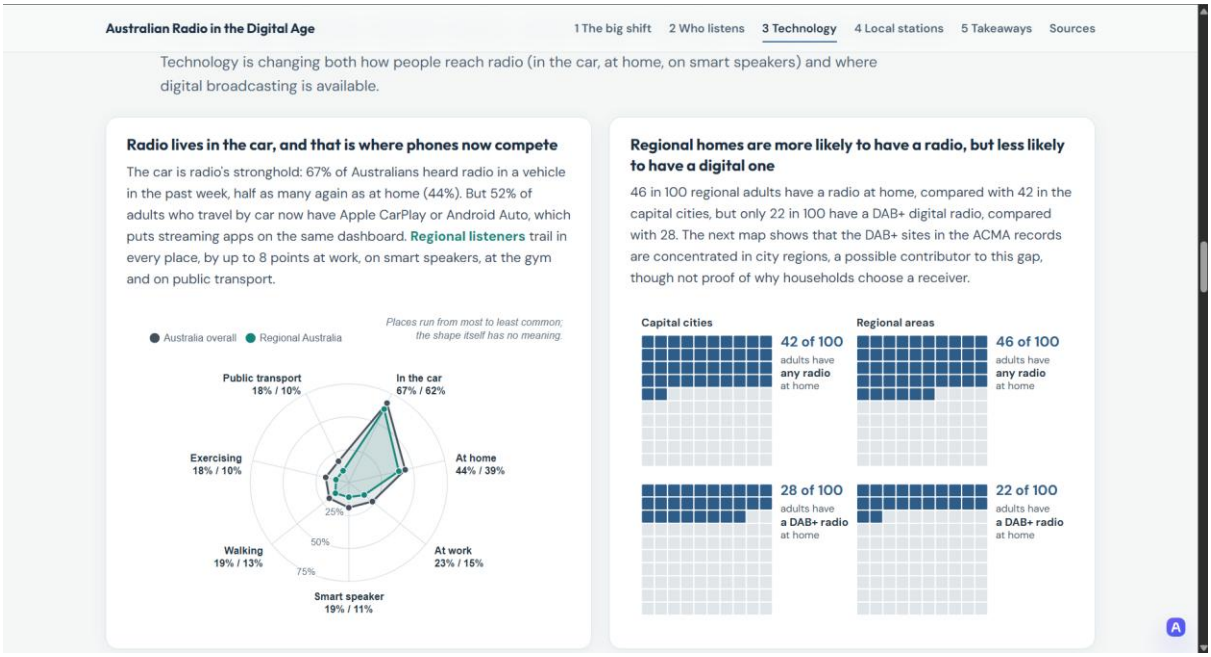
Paste a full-size screenshot of your DV II assignment from the URL here

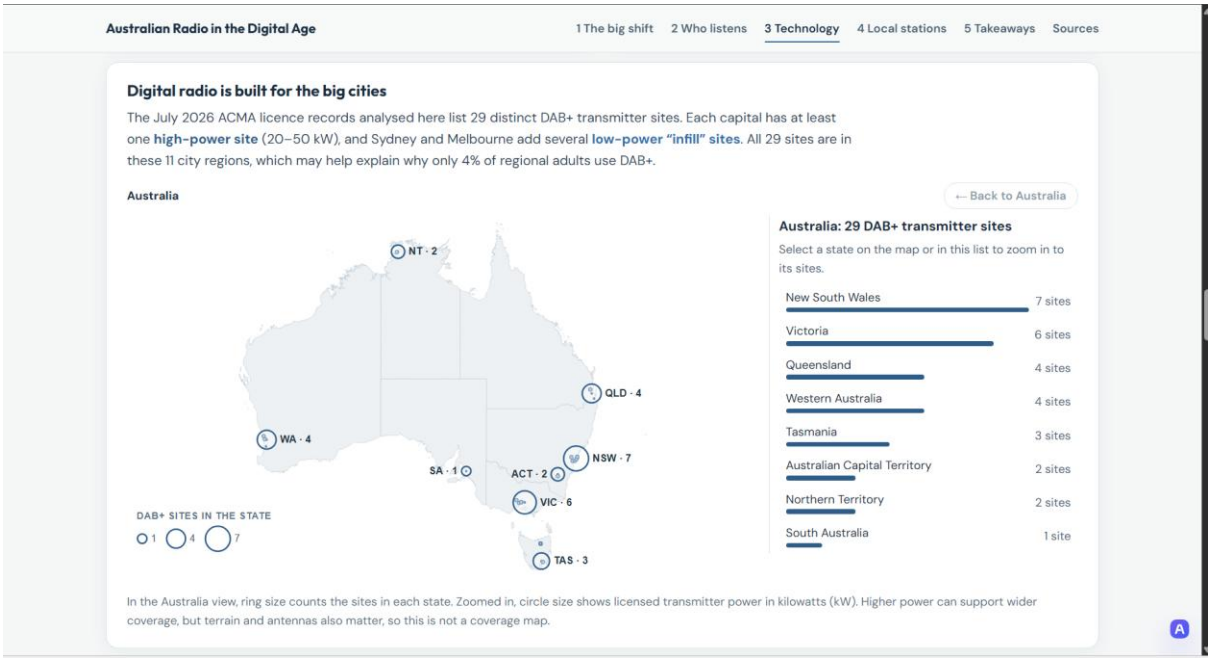


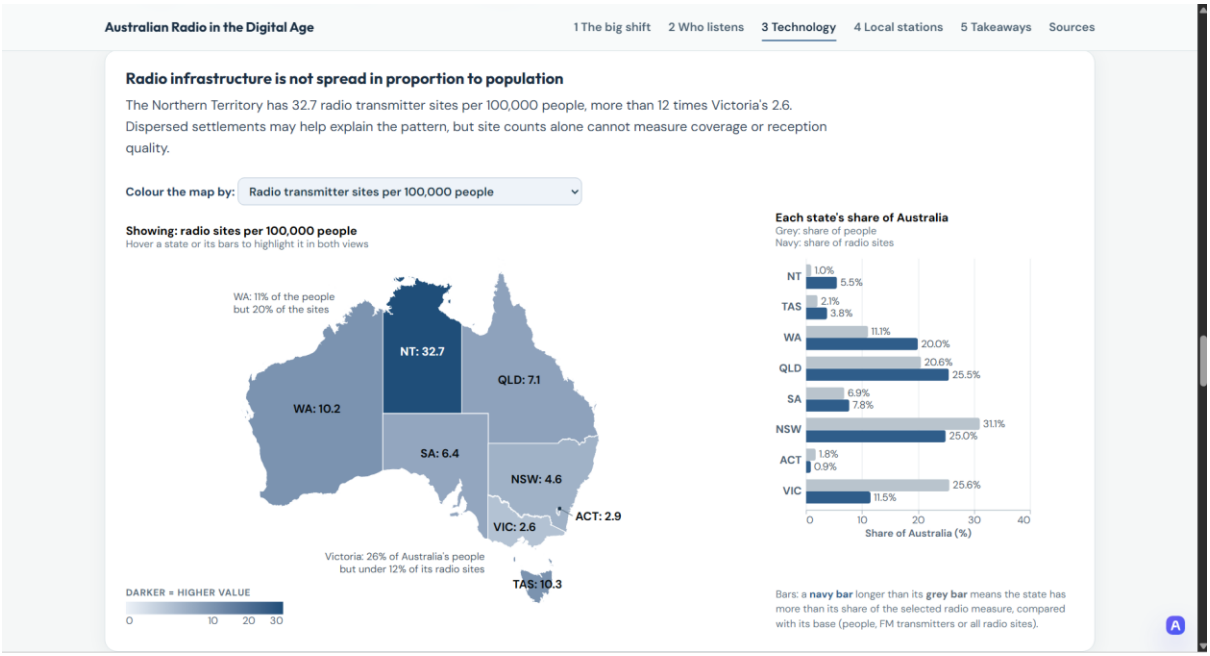

Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
Streaming overtook radio in 2025, but radio is still heard by 3 in 4 people
Since 2017, weekly streaming rose from 46% to 81% and podcasts from 10% to 45%, while radio slipped only from 85% to 77%. With 81% streaming and 77% on radio in 2026, at least about 58% of Australians used both each week: the survey shows overlap, not listeners switching.
CLICK TO HIGHLIGHT
Streaming audio
Any radio
Commercial radio
Podcasts
DAB+ digital radio
Streaming passes radio
Listened in the past week (%)
81%
77%
61%
45%
42%
2017 2018 2019 2020 2021 2022 2023 2024 2025 2026
Show audience:
Australia overall
▶ Play 2017 → 2026
Spotify leads YouTube Music by 18 points (47% vs 29%), and growth slowed in 2026
Spotify is used weekly by almost half of Australians (47%), and YouTube Music is a clear second at 29%, almost double Apple Music's 16%. Apple Music and Amazon Music grew fastest in 2025, then held steady in 2026. Three years is too short to say whether growth has stopped.
2024
2025
2026
Spotify
46%
48%
▲2
47%
▼1
YouTube Music
24%
29%
▲5
29%
no change
Apple Music
10%
16%
▲6
16%
no change
Amazon Music
6%
11%
▲5
11%
no change
SoundCloud
7%
9%
▲2
8%
▼1
USED WEEKLY (%)
0 10 20 30 40 50

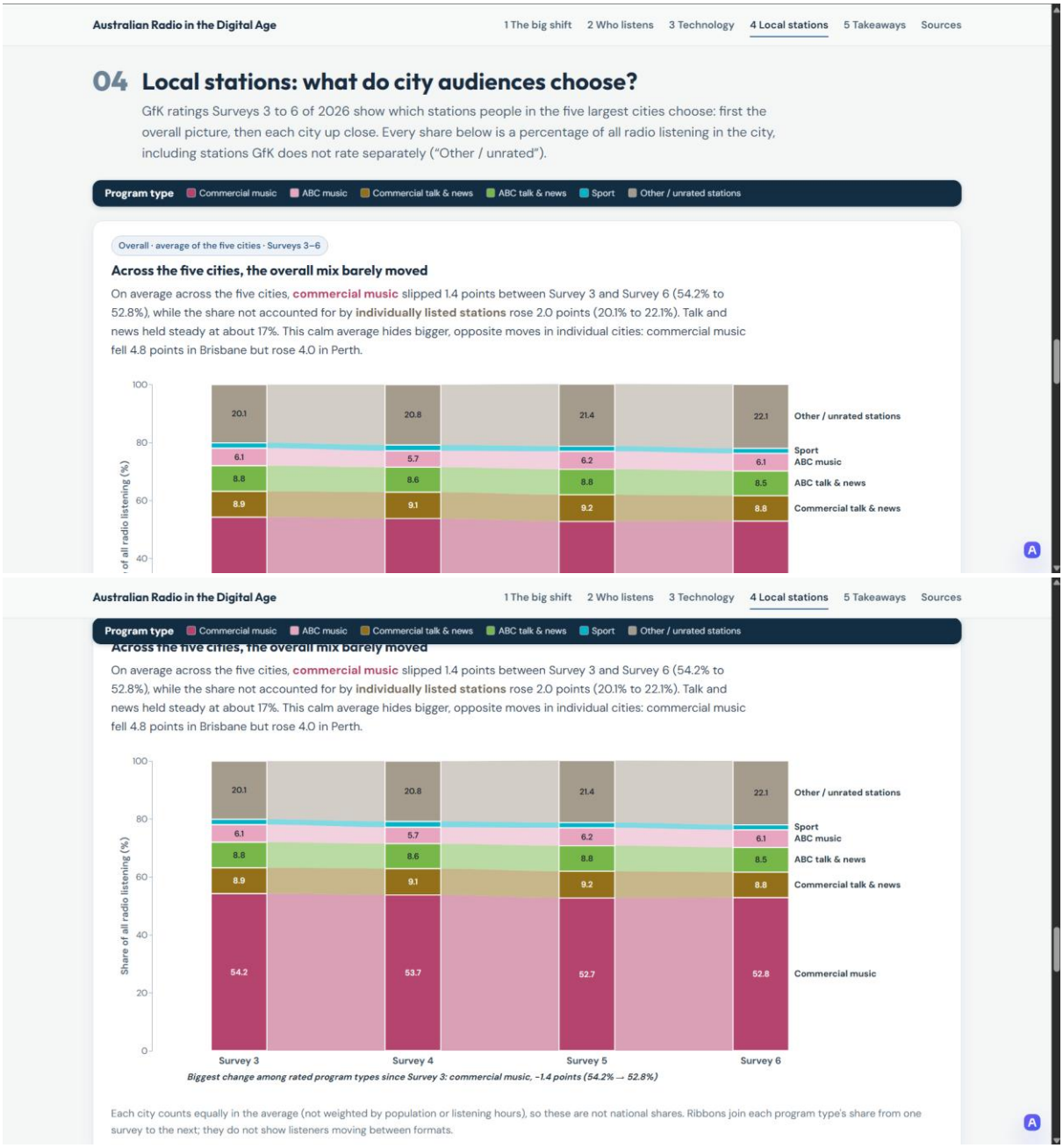

Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
02 Who listens: age and location matter
Weekly radio reach is highest among older Australians. FM is used more outside the capitals, while DAB+, streaming and podcasts are used more in the cities.
Young regional Australians have the lowest weekly radio reach
8 in 10 Australians aged 55+ listen to radio each week, in cities and regions alike. Among 10–24 year olds the gap is 10 points: 70% nationally but only 60% in regional areas. This is a 2026 snapshot, so it compares groups rather than showing change over time.
10–24
Gap: 10 points
60
70
25–54
Gap: 8 points
69
77
55+
Same reach: 80%
All ages
Gap: 5 points
72
77
Age
50 60 70 80
Weekly radio reach (%)
Australia overall
Regional Australia
Axis starts at 45%
A third of Australians aged 10+ listen to and watch podcasts
Podcast audio and video audiences overlap. In 2026, 55% of Australians aged 10+ consumed podcasts each month: 35% of all Australians both listened and watched, while only 5% watched without listening. Monthly podcast use is lower in regional Australia (47%).
Australia overall
Regional Australia
35%
5%
45%
15%
28%
4%
53%
15%
Listened 50%
Did not listen 50%
Listened 43%
Did not listen 57%
PODCAST BEHAVIOUR
Both
Listened only
Watched only
Neither


Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
Regional Australians stick with FM; city residents go digital
FM radio is 8 points more popular in regional areas (58% vs 50%), while DAB+ digital radio is 10 points more popular in the cities (14% vs 4%). Streaming and podcasts also lean towards the cities. Chapter 3 shows that DAB+ towers are concentrated in the capitals, which may help explain the gap.
More popular in capital cities
More popular in regional areas
FM radio
Online radio
Music streaming
Podcasts
DAB+ radio
Regional 58% vs capitals 50%
+8 points
Regional 10% vs capitals 7%
+3 points
-4 points
Regional 70% vs capitals 74%
-6 points
Regional 48% vs capitals 54%
-10 points
Regional 4% vs capitals 14%
-10
-5
0
+5
+10
Difference in weekly use (percentage points)
Why 14% here but 42% in the first chart? This chart uses the ACMA survey of adults, while the first chart uses Infinite Dial, which asks people aged 10+ a different question. Compare numbers within one survey, not across them.
03 How Australians reach radio, and where digital radio can operate

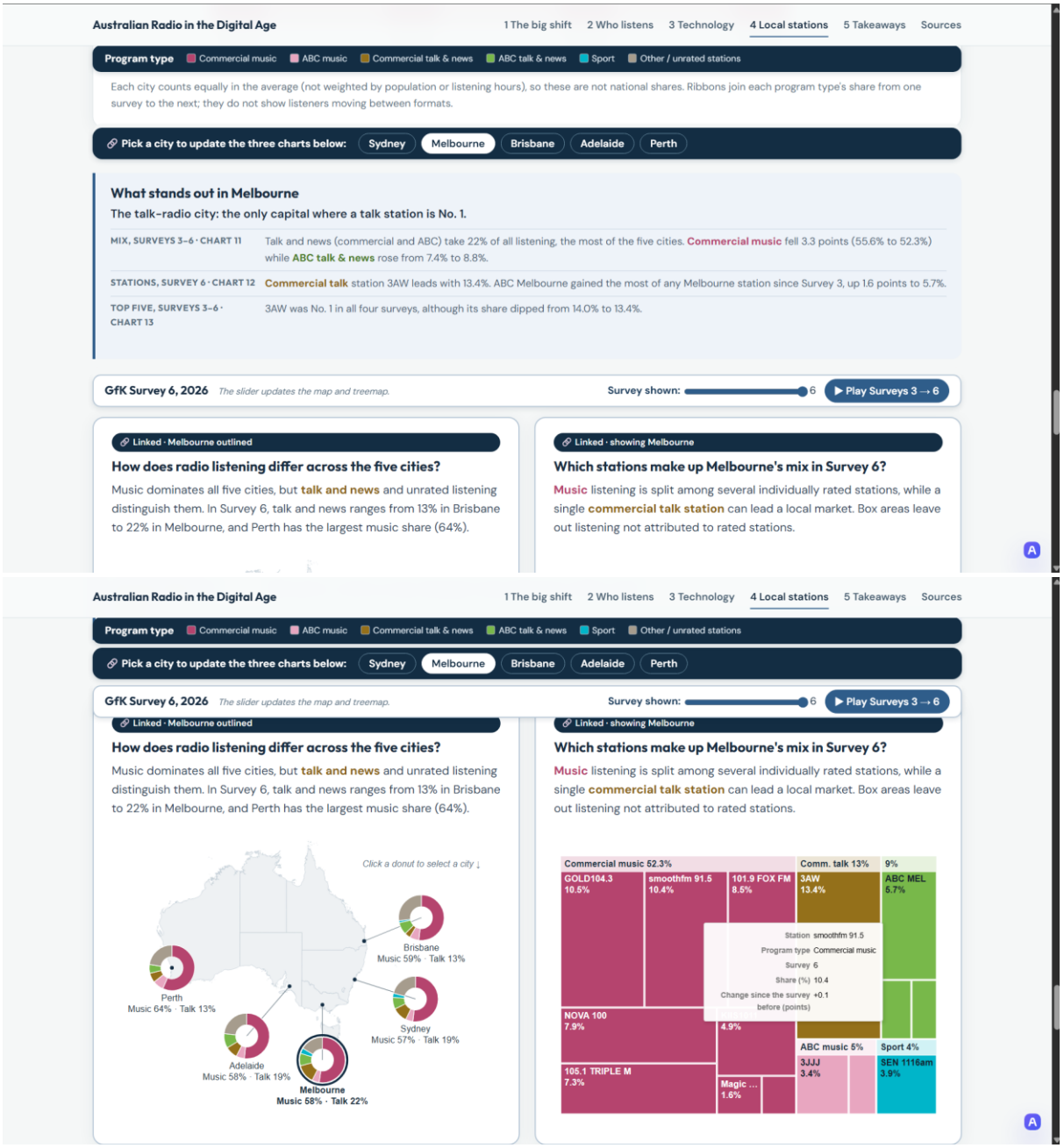

Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
Technology is changing both how people reach radio (in the car, at home, on smart speakers) and where digital broadcasting is available.
Radio lives in the car, and that is where phones now compete
The car is radio's stronghold: 67% of Australians heard radio in a vehicle in the past week, half as many again as at home (44%). But 52% of adults who travel by car now have Apple CarPlay or Android Auto, which puts streaming apps on the same dashboard. Regional listeners trail in every place, by up to 8 points at work, on smart speakers, at the gym and on public transport.
Australia overall
Regional Australia
Places run from most to least common; the shape itself has no meaning.
Public transport 18% / 10%
In the car 67% / 62%
At home 44% / 39%
At work 23% / 15%
Smart speaker 19% / 11%
Walking 19% / 13%
Exercising 18% / 10%
25%
50%
75%
Regional homes are more likely to have a radio, but less likely to have a digital one
46 in 100 regional adults have a radio at home, compared with 42 in the capital cities, but only 22 in 100 have a DAB+ digital radio, compared with 28. The next map shows that the DAB+ sites in the ACMA records are concentrated in city regions, a possible contributor to this gap, though not proof of why households choose a receiver.
Capital cities
Regional areas
42 of 100 adults have any radio at home
46 of 100 adults have any radio at home
28 of 100 adults have a DAB+ radio at home
22 of 100 adults have a DAB+ radio at home


Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
Digital radio is built for the big cities
The July 2026 ACMA licence records analysed here list 29 distinct DAB+ transmitter sites. Each capital has at least one high-power site (20–50 kW), and Sydney and Melbourne add several low-power "infill" sites. All 29 sites are in these 11 city regions, which may help explain why only 4% of regional adults use DAB+.
Australia
← Back to Australia
NT · 2
QLD · 4
WA · 4
SA · 1
ACT · 2
NSW · 7
VIC · 6
TAS · 3
DAB+ SITES IN THE STATE
1
4
7
Australia: 29 DAB+ transmitter sites
Select a state on the map or in this list to zoom in to its sites.
New South Wales 7 sites
Victoria 6 sites
Queensland 4 sites
Western Australia 4 sites
Tasmania 3 sites
Australian Capital Territory 2 sites
Northern Territory 2 sites
South Australia 1 site
In the Australia view, ring size counts the sites in each state. Zoomed in, circle size shows licensed transmitter power in kilowatts (kW). Higher power can support wider coverage, but terrain and antennas also matter, so this is not a coverage map.

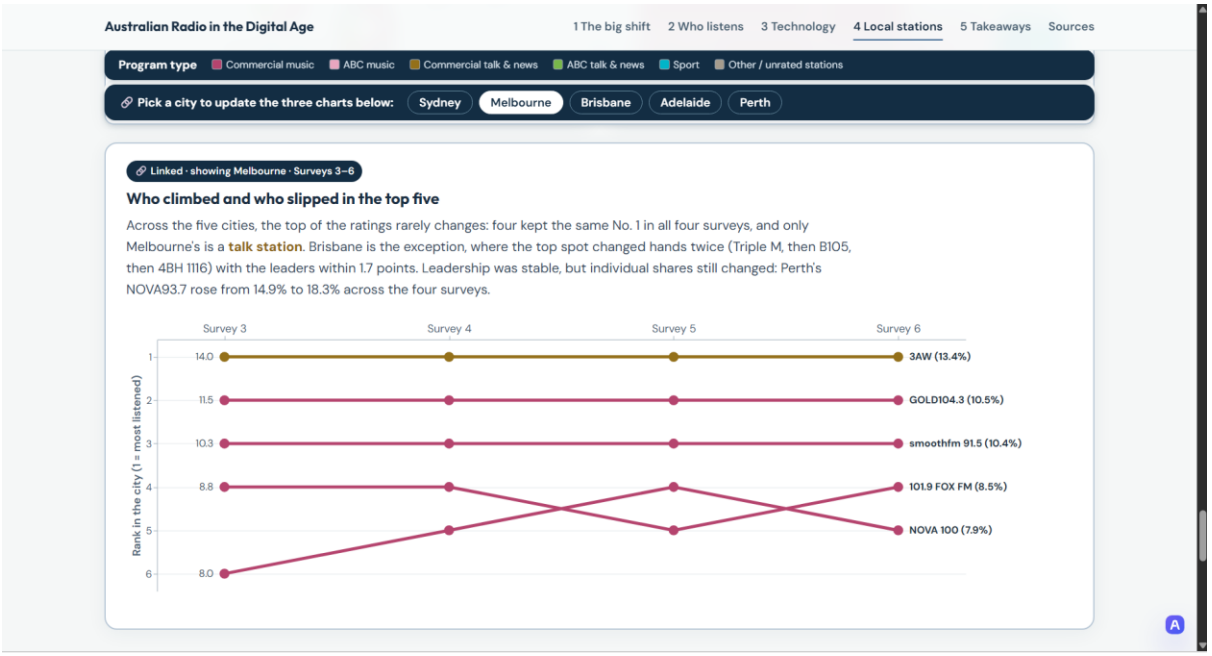

Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
Radio infrastructure is not spread in proportion to population
The Northern Territory has 32.7 radio transmitter sites per 100,000 people, more than 12 times Victoria's 2.6. Dispersed settlements may help explain the pattern, but site counts alone cannot measure coverage or reception quality.
Colour the map by: Radio transmitter sites per 100,000 people
Showing: radio sites per 100,000 people
Hover a state or its bars to highlight it in both views
WA: 11% of the people but 20% of the sites
NT: 32.7
QLD: 7.1
WA: 10.2
SA: 6.4
NSW: 4.6
ACT: 2.9
VIC: 2.6
Victoria: 26% of Australia's people but under 12% of its radio sites
TAS: 10.3
DARKER = HIGHER VALUE
0 10 20 30
Each state's share of Australia
Grey: share of people
Navy: share of radio sites
NT 1.0% 5.5%
TAS 2.1% 3.8%
WA 11.1% 20.0%
QLD 20.6% 25.5%
SA 6.9% 7.8%
NSW 31.1% 25.0%
ACT 1.8% 0.9%
VIC 25.6% 11.5%
0 10 20 30 40
Share of Australia (%)
Bars: a navy bar longer than its grey bar means the state has more than its share of the selected radio measure, compared with its base (people, FM transmitters or all radio sites).

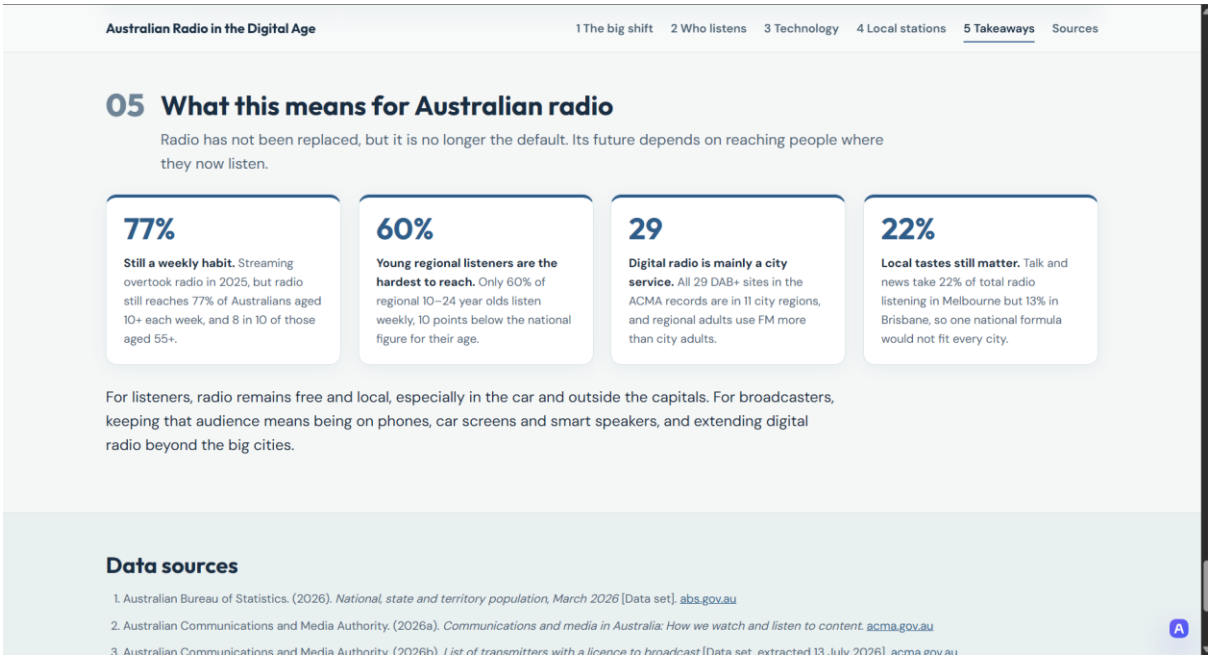

Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
04 Local stations: what do city audiences choose?
GfK ratings Surveys 3 to 6 of 2026 show which stations people in the five largest cities choose: first the overall picture, then each city up close. Every share below is a percentage of all radio listening in the city, including stations GfK does not rate separately ("Other / unrated").
Program type
Commercial music
ABC music
Commercial talk & news
ABC talk & news
Sport
Other / unrated stations
Overall · average of the five cities · Surveys 3–6
Across the five cities, the overall mix barely moved
On average across the five cities, commercial music slipped 1.4 points between Survey 3 and Survey 6 (54.2% to 52.8%), while the share not accounted for by individually listed stations rose 2.0 points (20.1% to 22.1%). Talk and news held steady at about 17%. This calm average hides bigger, opposite moves in individual cities: commercial music fell 4.8 points in Brisbane but rose 4.0 in Perth.
100
80
60
40
20.1
20.8
21.4
22.1
6.1
5.7
6.2
6.1
8.8
8.6
8.8
8.5
8.9
9.1
9.2
8.8
Other / unrated stations
Sport
ABC music
ABC talk & news
Commercial talk & news

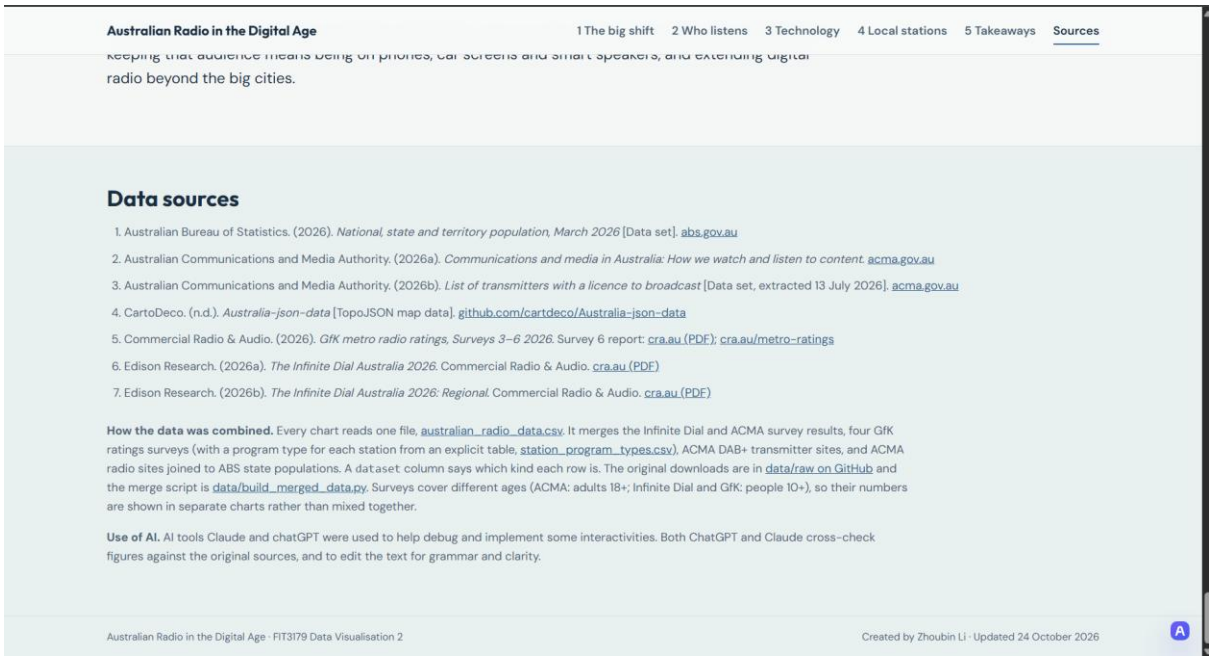
Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
Program type
Commercial music
ABC music
Commercial talk & news
ABC talk & news
Sport
Other / unrated stations
Across the five cities, the overall mix barely moved
On average across the five cities, commercial music slipped 1.4 points between Survey 3 and Survey 6 (54.2% to 52.8%), while the share not accounted for by individually listed stations rose 2.0 points (20.1% to 22.1%). Talk and news held steady at about 17%. This calm average hides bigger, opposite moves in individual cities: commercial music fell 4.8 points in Brisbane but rose 4.0 in Perth.
Share of all radio listening (%)
100
80
60
40
20
0
20.1
20.8
21.4
22.1
6.1
5.7
6.2
6.1
8.8
8.6
8.8
8.5
8.9
9.1
9.2
8.8
54.2
53.7
52.7
52.8
Other / unrated stations
Sport
ABC music
ABC talk & news
Commercial talk & news
Commercial music
Survey 3
Survey 4
Survey 5
Survey 6
Biggest change among rated program types since Survey 3: commercial music, −1.4 points (54.2% → 52.8%)
Each city counts equally in the average (not weighted by population or listening hours), so these are not national shares. Ribbons join each program type's share from one survey to the next; they do not show listeners moving between formats.


Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
Program type
Commercial music
ABC music
Commercial talk & news
ABC talk & news
Sport
Other / unrated stations
Each city counts equally in the average (not weighted by population or listening hours), so these are not national shares. Ribbons join each program type's share from one survey to the next; they do not show listeners moving between formats.
Pick a city to update the three charts below:
Sydney
Melbourne
Brisbane
Adelaide
Perth
What stands out in Melbourne
The talk-radio city: the only capital where a talk station is No. 1.
MIX, SURVEYS 3–6 · CHART 11
Talk and news (commercial and ABC) take 22% of all listening, the most of the five cities. Commercial music fell 3.3 points (55.6% to 52.3%) while ABC talk & news rose from 7.4% to 8.8%.
STATIONS, SURVEY 6 · CHART 12
Commercial talk station 3AW leads with 13.4%. ABC Melbourne gained the most of any Melbourne station since Survey 3, up 1.6 points to 5.7%.
TOP FIVE, SURVEYS 3–6 · CHART 13
3AW was No. 1 in all four surveys, although its share dipped from 14.0% to 13.4%.
GfK Survey 6, 2026
The slider updates the map and treemap.
Survey shown:
6
Play Surveys 3 → 6
Linked · Melbourne outlined
How does radio listening differ across the five cities?
Music dominates all five cities, but talk and news and unrated listening distinguish them. In Survey 6, talk and news ranges from 13% in Brisbane to 22% in Melbourne, and Perth has the largest music share (64%).
Linked · showing Melbourne
Which stations make up Melbourne's mix in Survey 6?
Music listening is split among several individually rated stations, while a single commercial talk station can lead a local market. Box areas leave out listening not attributed to rated stations.

Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
Program type
Commercial music
ABC music
Commercial talk & news
ABC talk & news
Sport
Other / unrated stations
Pick a city to update the three charts below:
Sydney
Melbourne
Brisbane
Adelaide
Perth
GfK Survey 6, 2026
The slider updates the map and treemap.
Survey shown:
6
Play Surveys 3 → 6
Linked · Melbourne outlined
How does radio listening differ across the five cities?
Music dominates all five cities, but talk and news and unrated listening distinguish them. In Survey 6, talk and news ranges from 13% in Brisbane to 22% in Melbourne, and Perth has the largest music share (64%).
Click a donut to select a city ↓
Brisbane
Music 59% · Talk 13%
Perth
Music 64% · Talk 13%
Sydney
Music 57% · Talk 19%
Adelaide
Music 58% · Talk 19%
Melbourne
Music 58% · Talk 22%
Linked · showing Melbourne
Which stations make up Melbourne's mix in Survey 6?
Music listening is split among several individually rated stations, while a single commercial talk station can lead a local market. Box areas leave out listening not attributed to rated stations.
Commercial music 52.3%
Comm. talk 13%
9%
GOLD104.3
10.5%
smoothfm 91.5
10.4%
101.9 FOX FM
8.5%
3AW
13.4%
ABC MEL
5.7%
Station smoothfm 91.5
Program type Commercial music
Survey 6
Share (%) 10.4
Change since the survey before (points) +0.1
NOVA 100
7.9%
4.9%
ABC music 5%
Sport 4%
105.1 TRIPLE M
7.3%
3JJJ
3.4%
SEN 1116am
3.9%
Magic …
1.6%


Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
Program type
Commercial music
ABC music
Commercial talk & news
ABC talk & news
Sport
Other / unrated stations
Pick a city to update the three charts below:
Sydney
Melbourne
Brisbane
Adelaide
Perth
Linked · showing Melbourne · Surveys 3–6
Who climbed and who slipped in the top five
Across the five cities, the top of the ratings rarely changes: four kept the same No. 1 in all four surveys, and only Melbourne's is a talk station. Brisbane is the exception, where the top spot changed hands twice (Triple M, then B105, then 4BH 1116) with the leaders within 1.7 points. Leadership was stable, but individual shares still changed: Perth's NOVA93.7 rose from 14.9% to 18.3% across the four surveys.
Survey 3
Survey 4
Survey 5
Survey 6
Rank in the city (1 = most listened)
1
2
3
4
5
6
14.0
11.5
10.3
8.8
8.0
3AW (13.4%)
GOLD104.3 (10.5%)
smoothfm 91.5 (10.4%)
101.9 FOX FM (8.5%)
NOVA 100 (7.9%)


Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
05 What this means for Australian radio
Radio has not been replaced, but it is no longer the default. Its future depends on reaching people where they now listen.
77%
Still a weekly habit. Streaming overtook radio in 2025, but radio still reaches 77% of Australians aged 10+ each week, and 8 in 10 of those aged 55+.
60%
Young regional listeners are the hardest to reach. Only 60% of regional 10–24 year olds listen weekly, 10 points below the national figure for their age.
29
Digital radio is mainly a city service. All 29 DAB+ sites in the ACMA records are in 11 city regions, and regional adults use FM more than city adults.
22%
Local tastes still matter. Talk and news take 22% of total radio listening in Melbourne but 13% in Brisbane, so one national formula would not fit every city.
For listeners, radio remains free and local, especially in the car and outside the capitals. For broadcasters, keeping that audience means being on phones, car screens and smart speakers, and extending digital radio beyond the big cities.
Data sources
1. Australian Bureau of Statistics. (2026). National, state and territory population, March 2026 [Data set]. abs.gov.au
2. Australian Communications and Media Authority. (2026a). Communications and media in Australia: How we watch and listen to content. acma.gov.au
3. Australian Communications and Media Authority. (2026b). List of transmitters with a licence to broadcast [Data set, extracted 13 July 2026]. acma.gov.au


Australian Radio in the Digital Age
1 The big shift
2 Who listens
3 Technology
4 Local stations
5 Takeaways
Sources
radio beyond the big cities.
Data sources
1. Australian Bureau of Statistics. (2026). National, state and territory population, March 2026 [Data set]. abs.gov.au
2. Australian Communications and Media Authority. (2026a). Communications and media in Australia: How we watch and listen to content. acma.gov.au
3. Australian Communications and Media Authority. (2026b). List of transmitters with a licence to broadcast [Data set, extracted 13 July 2026]. acma.gov.au
4. CartoDeco. (n.d.). Australia-json-data [TopoJSON map data]. github.com/cartdeco/Australia-json-data
5. Commercial Radio & Audio. (2026). GfK metro radio ratings, Surveys 3–6 2026. Survey 6 report: cra.au (PDF); cra.au/metro-ratings
6. Edison Research. (2026a). The Infinite Dial Australia 2026. Commercial Radio & Audio. cra.au (PDF)
7. Edison Research. (2026b). The Infinite Dial Australia 2026: Regional. Commercial Radio & Audio. cra.au (PDF)
How the data was combined. Every chart reads one file, australian_radio_data.csv. It merges the Infinite Dial and ACMA survey results, four GfK ratings surveys (with a program type for each station from an explicit table, station_program_types.csv), ACMA DAB+ transmitter sites, and ACMA radio sites joined to ABS state populations. A dataset column says which kind each row is. The original downloads are in data/raw on GitHub and the merge script is data/build_merged_data.py. Surveys cover different ages (ACMA: adults 18+; Infinite Dial and GfK: people 10+), so their numbers are shown in separate charts rather than mixed together.
Use of AI. AI tools Claude and chatGPT were used to help debug and implement some interactivities. Both ChatGPT and Claude cross-check figures against the original sources, and to edit the text for grammar and clarity.
Australian Radio in the Digital Age · FIT3179 Data Visualisation 2
Created by Zhoubin Li · Updated 24 October 2026